

FULL PROJECT LIFECYCLE ROADMAP

Quantifying Latent Policy Stance: Extracting Implicit Macroeconomic Sentiment from Central Bank Communications

5 Phases · 12 Milestones · Targeting JFE / RFS / ACL / EMNLP

Bird's-Eye View: Full Project Trajectory

The table below maps the complete research lifecycle. Each phase contains discrete milestones with defined entry and exit criteria. Phases 1–3 are sequential and build the core technical contribution; Phase 4 is the quality-assurance gate that determines publishability in a top-tier venue; Phase 5 governs manuscript construction and journal strategy.

Phase	Title	Milestones	Est. Time	Core Deliverable
PHASE 1	Data Engineering	M1–M2	4–6 weeks	FOMC corpus + HF tick data pipeline; event-window labels
PHASE 2	Baseline Models	M3	2–3 weeks	LM lexicon + FinBERT benchmarks; performance ceiling defined
PHASE 3	Advanced ML	M4–M6	6–10 weeks	LoRA fine-tuning; embedding extraction; stance index
PHASE 4	Robustness & OOS	M7–M9	4–6 weeks	Walk-forward CV; ablations; economic significance tests
PHASE 5	Manuscript & Submit	M10–M12	8–12 weeks	Paper writing; revisions; journal targeting & submission

Total estimated project duration from data acquisition to manuscript submission: 24–37 weeks (6–9 months), assuming full-time research effort. Part-time effort scales proportionally. The single largest risk to timeline is data acquisition in Phase 1 (licensing high-frequency tick data) and reviewer revision cycles in Phase 5.

PHASE 1 · FOUNDATION & DATA ENGINEERING

Constructing the corpus, tick-data pipeline, and event-window labels that constitute the empirical substrate of the entire project.

MILESTONE 1

FOMC Corpus Acquisition, Parsing & Versioning

EST. DURATION

2–3 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Write scrapers for federalreserve.gov targeting: (a) post-meeting Statements, (b) Meeting Minutes PDFs, (c) Press Conference transcripts (video-derived transcripts from 2011+). • Parse PDFs to clean UTF-8 text (pdfplumber / pypdf2); strip headers, footers, page numbers, and legal boilerplate. • Build a document registry: each record stores document_type, fomc_date, release_datetime_UTC, raw_text, and a SHA-256 content hash. • Implement paragraph- and sentence-level segmentation using spaCy with economic-domain tokenization overrides (e.g., do not split '2.25%' or 'Q3 2022'). • Store each chunk with metadata: doc_id, chunk_id, chunk_type (paragraph/sentence), position_in_doc, release_datetime_UTC. • Version-control the corpus in a reproducible manner (DVC or git-annex) so any reviewer can reconstruct the exact dataset. • Manual audit: randomly sample 50 chunks per document type	• Corpus contains ≥ 240 distinct FOMC events spanning Jan 2005–present with zero duplicates (verified by SHA-256 hash). • All three document types represented with correct release timestamps accurate to the minute (cross-checked against FOMC historical calendar). • Chunk registry contains $\geq 25,000$ labeled chunks with complete metadata; $< 1\%$ segmentation errors in manual audit. • Pipeline is fully automated and re-runnable from a single script with no manual intervention. • Corpus passes a reproducibility check: a second researcher running the pipeline obtains an identical SHA-256 corpus hash.	• Release timestamp error: using the FOMC meeting date rather than the exact minute of document release invalidates all event-window calculations. Verify against press release timestamps on federalreserve.gov, not calendar dates. • Parsing artifacts: PDF tables of economic projections, charts, and footnotes will corrupt text if not stripped. Failure to clean these poisons downstream embeddings. • Data leakage in corpus construction: inadvertently including revised or post-hoc released documents (e.g., transcript corrections) as if they were real-time. Use only information available at original release time. • Ignoring document type heterogeneity: treating a 400-word statement and a 6,000-word minutes document identically will distort chunk-level analysis.

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
and verify segmentation and metadata accuracy.		

MILESTONE 2**High-Frequency Tick Data Pipeline & Event-Window Label Construction**

EST.
DURATION
2–3 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• License or acquire 2-Year US Treasury Note futures (TU) tick data from CME DataMine, Refinitiv Tick History, or Bloomberg B-PIPE; minimum resolution: 1-second bid/ask. • Acquire S&P 500 E-mini futures (ES) tick data from the same source and time range. • Build an event-window extraction function: for each FOMC release_datetime_UTC, extract the price series over [t–15min, t+60min] at 1-minute OHLCV resolution. • Compute primary label Δy_2 : signed basis-point change in 2-Year yield from the last pre-release mid-price (t–1min) to the peak post-release mid-price within [t, t+30min]. • Compute secondary labels: (a) ΔES: % change in S&P 500 futures; (b) $\Delta slope$: change in 2s10s curve slope; (c) Δvol: change in MOVE index if available. • Match labels to chunks: assign the event-window label of the containing FOMC release to all text chunks from that release. • Document every data source, licensing terms, and any gaps or	• Every FOMC event in the corpus has a matched, non-null Δy_2 label (allow <5% missingness for documented data gaps). • Label distribution is inspected: mean ≈ 0 basis points (no unconditional policy bias), std dev consistent with historical literature (~5–8 bp per event). • Event-window plots for ≥ 10 randomly sampled events visually confirm price impact is isolated in the post-release window. • All labels stored with event_id, release_datetime_UTC, Δy_2 _bp, ΔES _pct, data_source, and a quality_flag for any anomalous events. • Data appendix drafted and ready for Methods section.	• Look-ahead bias in label construction: the single most fatal error for a top journal. The label must be computed from prices after the release, using only the timestamp of the release as the boundary. Any use of end-of-day yields or daily closing prices conflates the FOMC signal with subsequent news. • Timezone handling errors: FOMC statements are released in US/Eastern time; tick data is often stored in UTC. A timezone mismatch of even one hour places the pre-release price inside the post-release window. • Overnight/weekend events: some FOMC events occur during emergency inter-meeting actions with atypical market conditions. These must be flagged separately, not silently included, as they confound the model. • Survivorship and availability bias: if using a vendor that only covers liquid periods, gaps in the 2008–2009 crisis period may systematically drop the most volatility-informative events.

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
missing events in a data appendix.		

PHASE 2 · BASELINE ESTABLISHMENT

Running lexicon and early-transformer benchmarks to quantify the performance gap that motivates the advanced architecture.

MILESTONE 3

Lexicon & FinBERT Baselines: Quantifying the Performance Ceiling

EST.
DURATION
2–3 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
- Implement LM lexicon scoring: for each text chunk, compute (a) LM_neg: fraction of negative-category tokens, (b) LM_pos: fraction of positive tokens, (c) LM_net = LM_pos – LM_neg. - Implement a custom Hawkish-Dovish word list baseline: construct a small domain-specific lexicon (≈200 terms per class: 'tighten', 'inflation', 'overshoot' vs. 'accommodate', 'slack', 'patient') and compute an HD_net score. - Run FinBERT zero-shot inference: pass each chunk through FinBERT and record positive/negative/neutral probability as a continuous score. - Run FinBERT on a hawkishness framing: prompt-engineer the classification head by fine-tuning FinBERT on a small manually annotated HD sample (≈500 chunks labeled	- Baseline table is complete with all metrics (r, p, directional accuracy, MAE) for ≥4 baseline methods. - Results are statistically characterized: 95% confidence intervals on correlations via bootstrap (N=10,000 iterations). - A clear performance gap is documented: LM and FinBERT baselines show statistically significant but economically modest correlation with Δy_2 (expected $r \approx 0.10$–0.25 based on literature). - Code is modular and the baseline pipeline can be re-run deterministically given the corpus.	- P-hacking the baseline: do not tune hyperparameters of baseline models to minimize their performance and maximize the apparent gap. Use standard, published configurations. Reviewers will test this. - Aggregation mismatch: chunk-level scores must be aggregated to event-level before correlating with a single event-level Δy_2. Use a documented, theory-consistent aggregation (e.g., mean, max-abs, or statement-only chunks). - Ignoring multiple testing: if you compute correlations for 20 different baseline variants, apply a Bonferroni or Benjamini-Hochberg correction. Uncorrected p-values at this stage will haunt the paper in review. - Using the full sample for baseline evaluation before defining the train/test split: this

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hawk/dove/neutral by domain experts). • For each baseline, compute correlation (Pearson r, Spearman ρ) between the scalar score and Δy_2 across all events. • Compute directional accuracy (sign of score vs. sign of Δy_2) and MAE in basis points. • Run a Mincer-Zarnowitz regression of Δy_2 on each baseline score to test for forecast rationality. • Produce a baseline comparison table that will become Table 1 in the paper.		introduces implicit in-sample selection. Define the time-based split in M3 and use it consistently through all phases.

PHASE 3 · ADVANCED ML ARCHITECTURE & FINE-TUNING

Implementing the LoRA fine-tuning pipeline, extracting latent embeddings, and constructing the continuous hawkish-dovish stance index.

MILESTONE 4

Infrastructure Setup & Experiment Tracking

EST. DURATION

1–2 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Provision GPU compute: A100 80GB (recommended) or 2×A100 40GB via cloud provider (Lambda Labs, RunPod, AWS p4d). Budget ≈\$200–600 for full fine-tuning run. • Set up experiment tracking: MLflow or Weights & Biases for logging all hyperparameters, loss curves, and evaluation metrics per run.	• Training environment is fully documented and reproducible: a second researcher following the README can reproduce results within $\pm 1\%$ on all metrics. • Temporal split is locked and version-controlled: the split dates are hardcoded and cannot be adjusted post-hoc. • MLflow / W&B dashboard shows experiment tracking active and logging correctly.	• Data leakage through the train/test split: this is the most common and most fatal error in ML finance papers. Any parameter tuning, feature selection, or normalization that uses test-set statistics constitutes look-ahead bias and will cause a rejection at JFE/RFS. • Non-reproducible experiments: if you cannot reproduce your

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Implement reproducible training: fix all random seeds (Python, NumPy, PyTorch, Transformers); log environment (CUDA version, library versions, hardware) to a requirements.txt and environment.yml. • Implement a strict temporal train/validation/test split: Train = 2005–2018, Validation = 2019–2020, Test = 2021–present. Crucially, no information from validation or test periods may influence model selection or hyperparameter choice. • Define the regression head: a single linear layer on the [CLS]-equivalent token (or mean-pool of final hidden states) projecting from <code>d_model</code> to 1 scalar output. • Implement evaluation harness: for every model checkpoint, compute $r(\Delta y_2)$, directional accuracy, and MAE on the held-out validation set.	• Infrastructure cost estimate is documented for Methods section.	own results with a fixed seed, reviewers conducting replication checks will obtain different numbers, which is grounds for rejection at top venues. • Compute cost miscalculation: underestimating GPU-hours leads to rushed training runs with sub-optimal hyperparameters. Budget conservatively.

MILESTONE 5**LoRA Fine-Tuning: Regression on High-Frequency Market Labels**

EST.
DURATION
3–5 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Load base model: Llama-3-8B-Instruct or Mistral-7B-Instruct-v0.3 in 4-bit NF4 quantization (bitsandbytes) for memory efficiency. • Configure LoRA via HuggingFace PEFT: target modules = <code>['q_proj', 'v_proj', 'k_proj', 'o_proj']</code>; rank $r \in \{8, 16, 32\}$	• Best checkpoint achieves validation Pearson $r > 0.35$ (significantly above best baseline of ~ 0.25); if not, investigate data pipeline before proceeding. • Null model check passes: shuffled-label model achieves $r < 0.05$ on validation set. • Hyperparameter sweep is complete: best configuration	• Overfitting to the training period: the FOMC has changed communication style dramatically across the sample. A model that memorizes 2010-era language patterns will fail post-2021. Monitor the gap between train and validation MSE; a gap $> 2\times$ signals overfitting. • Prompt template contamination: if the input prompt includes any information

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
(hyperparameter sweep); $\alpha = 2r$; dropout = 0.05. - Fine-tune on train split with MSE loss, AdamW optimizer ($\text{lr} = 1\text{e-}4$, weight decay = 0.01), cosine LR schedule with warmup, gradient clipping ($\text{max_norm}=1.0$). - Train for up to 10 epochs with early stopping on validation MSE (patience = 3 epochs); save best checkpoint by validation correlation r. - Run hyperparameter sweep over: LoRA rank, learning rate, chunk aggregation strategy (mean vs. max-abs vs. statement-only), and input prompt template. - Compare Llama-3-8B vs. Mistral-7B on identical setups; document architectural differences relevant to economic text. - Conduct a 'null model' sanity check: fine-tune the same architecture on randomly shuffled labels. The shuffled model must show $r \approx 0$ on the validation set; if not, there is a data leakage bug.	documented with full W&B run log. All checkpoints and training configs saved to version control; training can be reproduced exactly.	about the macroeconomic context available at the time of release (e.g., 'Given that CPI was 8.5% at this meeting...'), you have introduced look-ahead bias. The model input must contain text only. - Selecting hyperparameters on test performance: even once. Any decision informed by test-set performance invalidates the out-of-sample claim. The test set is touched exactly once, at the final reporting stage. - Insufficient LoRA rank ablation: reviewers will ask whether a simpler $r=4$ model works just as well. Run the ablation now.

MILESTONE 6**Embedding Extraction, Latent Stance Index & Interpretability**

EST. DURATION

2–3 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
- Detach regression head; set model to eval mode; disable gradient computation. - Forward pass all N corpus chunks through the fine-tuned model; extract final-layer hidden	- PC1 explains $\geq 15\%$ of total embedding variance and has Pearson $r > 0.30$ with Δy_2 on the training split. - UMAP visualization shows coherent structure: hawkish and	PCA fitted on full dataset including test split: any dimensionality reduction or normalization that uses test-period observations contaminates the OOS

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states; store as $E \in \mathbb{R}^{(N \times 4096)}$ in memory-mapped format (numpy .npy or HDF5). • Fit PCA on training-split embeddings only; apply the training-derived PCA transform to all splits (no leakage). • Compute the latent stance score $s(x) = e(x)^T v_1$ for every chunk; verify that PC1 loading vector aligns with Δy_2 by computing $\text{corr}(s, \Delta y_2)$ on train split. • Apply UMAP (<code>n_components=2</code>, <code>n_neighbors=15</code>, <code>min_dist=0.1</code>, <code>metric='cosine'</code>) for visualization; produce scatter plot colored by Δy_2, document type, and calendar year. • Aggregate chunk-level scores to document-level: produce a time series of event-level hawkishness scores across the full 2005–present sample. • Conduct interpretability analysis: for the top-10 most hawkish and top-10 most dovish chunks by $s(x)$, extract and examine the text to confirm face validity. • Compute cosine similarity between embedding of each chunk and hand-crafted 'anchor' sentences ('the labor market remains very tight', 'we can afford to be patient').	dovish clusters are visually separable; temporal trajectory is economically interpretable (e.g., 2022 tightening cycle clusters distinctly). • Face validity check passes: top hawkish chunks contain language about inflation persistence, labor market tightness, rate increases; top dovish chunks reference accommodation, patience, downside risks. • Full embedding matrix is stored and versioned; PCA/UMAP transforms are serialized (pickle/joblib) for reproducible re-use.	evaluation. Fit all transforms on training data only. • Interpreting PC1 as 'hawkishness' without empirical validation: the label 'hawkish-dovish' must be earned by demonstrating correlation with Δy_2, not assumed. Do not name the dimension before verifying it. • Cherry-picking the interpretability examples: the face validity check must use the top chunks by objective ranking, not manually selected examples. Reviewers will ask to see random samples, not curated ones.

PHASE 4 · ROBUSTNESS & OUT-OF-SAMPLE TESTING

The quality-assurance gate that separates publishable research from an overfitted ML exercise. This phase determines whether the paper survives peer review.

MILESTONE 7**Walk-Forward Cross-Validation & OOS Performance**

EST. DURATION

2–3 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Implement a walk-forward (expanding-window) cross-validation scheme with a minimum 5-year training window and 1-year evaluation windows, stepping forward quarterly. • Re-train (or re-adapt via LoRA) the model at each walk-forward fold on the expanding training window; evaluate on the subsequent 1-year OOS window. • Compute the full battery of OOS metrics for each fold: Pearson r, Spearman ρ, directional accuracy, MAE, and the Diebold-Mariano test statistic against each baseline. • Report mean and standard deviation of all metrics across folds, not just the best fold. • Plot the time series of the model's Δy_2 prediction vs. actual Δy_2 across the full OOS period. • Conduct the final hold-out test: run inference on the locked test set (2021–present) exactly once and record all metrics. • Apply Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors to all OOS correlation tests.	• OOS Pearson r on final test set ≥ 0.35 and statistically significantly greater than best baseline (DM test $p < 0.05$ with HAC SEs). • Walk-forward results are stable across time: no single fold dominates the aggregate performance; the model does not rely on a single unusual event (e.g., post-COVID tightening) for its outperformance. • Final test set evaluation has been run exactly once; this is documented in the lab notebook. • Full walk-forward results table is ready for the paper's Appendix.	• Multiple testing on the final test set: running the test evaluation iteratively and stopping when you get a good result is the academic equivalent of p-hacking. The test set is used once, at the very end, for reporting purposes only. • Reporting only the best fold: top journals expect walk-forward average performance, not cherry-picked results. Present the full distribution across folds. • Ignoring transaction costs and market impact in any performance claim: if the paper makes any statement about trading profitability, this must net of realistic transaction costs and bid-ask spreads. • Correlation without economic significance: a Pearson r of 0.40 may be statistically significant but economically trivial if the MAE is 12 basis points and the average event impact is 4 bp. Distinguish statistical from economic significance throughout.

MILESTONE 8**Ablation Studies: Isolating the Source of Performance Gains**

EST. DURATION

1–2 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Ablation 1 — Base model size: compare 7B/8B vs. 1B parameter model (Llama-3.2-1B) under identical LoRA setup. Does scale matter? • Ablation 2 — LoRA rank: test $r \in \{4, 8, 16, 32, 64\}$. Plot validation r vs. rank to identify the performance plateau. • Ablation 3 — Pre-training domain: compare the instruction-tuned LLM against a vanilla BERT-base fine-tuned on the same regression task. Isolates the value of LLM-scale pre-training. • Ablation 4 — Supervision signal: compare (a) continuous Δy_2 regression vs. (b) discretized hawk/dove/neutral 3-class classification. Quantifies the information loss from categorization. • Ablation 5 — Document type: compare models trained on Statements only, Minutes only, Press Conferences only, and the full combined corpus. • Ablation 6 — Embedding layer: compare extracting embeddings from the last layer vs. second-to-last layer vs. a weighted average of last 4 layers. • Summarize all ablations in a structured table for the paper.	• All 6 ablations are complete with OOS metrics on the validation set. • A clear narrative emerges: each ablation identifies which design choices contribute meaningfully vs. marginally to performance. • The winning configuration is validated as the most parsimonious: if $r=8$ performs identically to $r=32$, use $r=8$ and report it as such.	• Conducting ablations after seeing test results and selecting ablations that make the model look good: all ablations must be pre-specified before any test-set exposure. Post-hoc ablation selection is a form of result mining. • Failure to include a vanilla BERT ablation: JFE/RFS reviewers will almost certainly ask 'why not just use FinBERT fine-tuned on your labels?' If you haven't answered this, the paper will be returned with a major revision request. • Reporting ablations only on in-sample performance: ablations must be evaluated on the same held-out validation fold used for model selection, not on the training set.

MILESTONE 9 Economic Significance, Placebo Tests & Alternative Explanations	EST. DURATION 1–2 weeks
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SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Economic significance test: construct a hypothetical long-short yield futures portfolio based on the model's directional signal; compute Sharpe ratio, maximum drawdown, and annualized return gross of transaction costs. Report alongside statistical metrics. • Placebo test 1 — Random text: run the model on randomly sampled non-FOMC text (e.g., Federal Reserve research publications not associated with policy decisions). The model must not produce significant Δy_2 predictions. • Placebo test 2 — Pre-event window: test whether the model's score on the pre-event window [t-60min, t-10min] predicts the subsequent yield move. It must not, or you have a reverse-causality problem. • Placebo test 3 — Macro surprise control: regress Δy_2 on both the model's stance score and a measure of macroeconomic surprise (e.g., Bloomberg consensus surprise for payrolls, CPI) released concurrently. Document that the model's contribution is incremental to macro surprise. • Alternative explanation test: verify that model performance is not driven by meeting-day fixed effects, time-of-day patterns, or seasonality by including these controls in multivariate regressions. • Heterogeneity analysis: does model performance differ across (a) tightening vs. easing cycles, (b) high vs. low macroeconomic uncertainty (VIX regimes), (c) statement type?	• All three placebo tests pass: model shows no predictive power on random text or pre-event windows. • Model contribution is statistically significant controlling for macro surprise (incremental $r^2 > 0$ at $p < 0.05$). • Economic significance section is drafted with the hypothetical portfolio analysis and appropriate caveats about transaction costs. • Heterogeneity results are summarized and provide additional economic insight (e.g., model performs better in high-uncertainty regimes).	• Claiming tradability without transaction cost netting: any statement that the model can generate alpha must account for bid-ask spreads in Treasury futures (typically 1/32 or ~0.03% for TU contracts) and market impact. Gross Sharpe ratios without this caveat will be flagged immediately. • Ignoring the macro surprise confound: the most common and most damaging alternative explanation a referee will raise is that the model is simply detecting whether an unexpected macro surprise occurred (which would move both yields and tone). This must be directly addressed. • Interpreting model predictions on unlabeled periods as validation: placebo tests require genuine null hypotheses with known-zero ground truth, not just low-Δy_2 periods.

PHASE 5 · MANUSCRIPT PREPARATION & SUBMISSION

Translating the technical results into a journal-quality paper, navigating the submission process, and managing the revision cycle.

MILESTONE 10

First Draft: Manuscript Structure & Core Sections

EST. DURATION
3–4 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Write the paper in the standard empirical finance structure: Abstract (150 words), Introduction (3–4 pages), Related Literature (2–3 pages), Data (2–3 pages), Methodology (3–4 pages), Results (4–5 pages), Robustness (2–3 pages), Conclusion (1–2 pages). • Introduction must follow the 'funnel' structure: (1) big-picture motivation, (2) specific research gap, (3) what this paper does, (4) main findings stated with numbers, (5) contribution to literature, (6) roadmap. • Methodology section must be self-contained and technically precise: a replication-capable reader must be able to reconstruct the pipeline from the methods section plus the appendix alone. • Results section: lead with the main table (OOS performance vs. baselines). Every table and figure must be interpretable standalone via caption. • Create all production-quality tables and figures: use consistent notation, at least 300 dpi for figures, and journal-standard table formatting (no	• Complete first draft of all sections exists, including all tables and figures in near-final form. • Abstract accurately summarizes the paper's contribution and main numerical findings. • Every empirical claim in the text is cross-referenced to a specific table, figure, or appendix item. • Technical appendix is complete and sufficient for replication.	• Burying the main result: the primary OOS correlation and comparison to baselines must appear in the abstract and early in the introduction with actual numbers. Papers that hide their main result behind 20 pages of methodology do not survive desk review at JFE. • Overclaiming contribution: top journals are particularly sensitive to papers that claim to 'solve' a problem or assert their method is universally superior. Frame contributions precisely and conditionally. • Undefined notation: every mathematical symbol must be defined at first use. Inconsistent notation across sections is a common referee complaint that signals careless writing. • Orphaned tables: every table and figure must be referenced in the main text and its economic interpretation discussed. Tables that appear without narrative explanation will receive referee criticism.

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
vertical lines, horizontal rules only for header and footer). • Write a Data Availability Statement specifying which data sources are licensed vs. public and how to access them. • Draft a technical appendix covering: full model architecture specifications, hyperparameter tables, walk-forward fold details, full ablation results, and any proofs.		

MILESTONE 11

Revision Cycles, Co-Author & Seminar Feedback

EST. DURATION
2–4 weeks

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Internal review pass: read the full manuscript aloud to identify unclear passages; verify every number in the text matches the corresponding table/figure. • Co-author / advisor review: share the draft with collaborators or mentors; incorporate substantive feedback on framing, methodology critique, and contribution claims. • Seminar presentation: present the paper at a departmental workshop, working group, or conference (NBER, SFS Cavalcade, AFA, ACL, EMNLP). Collect and categorize audience questions by type: (a) data/methodology, (b) interpretation, (c) contribution, (d) robustness. • Address seminar questions systematically: for each question, either (a) run the requested analysis and add to the paper, (b) add a footnote or appendix	• All co-author/advisor substantive comments have been addressed or formally documented as out-of-scope. • Seminar feedback has been categorized and addressed; no unaddressed methodological objection remains open. • Replication package is complete and tested: a clean environment running the README produces all main tables. • Paper is within the target journal's page limit and formatting requirements.	• Skipping seminar presentation: the seminar is a free simulation of peer review. Common seminar objections (especially about identification and alternative explanations) are exactly what referees will raise. Papers that skip this step are systematically less polished at submission. • Incomplete replication package: JFE and RFS now require code and data replication packages. Incomplete packages cause editorial desk rejections or conditional acceptances that stall for months. • Responding to feedback by adding text without removing old text: papers that accumulate responses to every seminar question without editing for coherence become internally inconsistent and overlong.

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section explaining why the analysis is not required, or (c) document why the question is out of scope. • Final pre-submission checklist: abstract ≤150 words, all tables have t-stats/standard errors, all figures are labeled, references are complete and in journal format, paper is within journal page limit. • Prepare supplemental replication package: code repository (GitHub or journal dataverse), data construction scripts, and a README that enables a third party to reproduce all tables.		

MILESTONE 12 Journal Targeting, Submission & Revision Management	EST. DURATION 3–6 weeks initial; ongoing
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Journal	Field	Accept%	IF	Strategic Notes
Journal of Financial Economics (JFE)	Finance	~12%	Very High	Best fit if empirical finance framing dominates; requires strong economic mechanism story
Review of Financial Studies (RFS)	Finance	~9%	Very High	Similar to JFE; slightly more receptive to methods-driven papers
Journal of Finance (JF)	Finance	~8%	Very High	Most prestigious; requires extremely broad appeal across finance subfields
Management Science (FinNLP Track)	Finance/OR	~15%	High	More receptive to ML-heavy methodologies; faster review cycle
Journal of Financial & Quantitative Analysis (JFQA)	Finance	~18%	High	Strong for empirical methods papers; good fallback from top-3
Computational Linguistics / ACL / EMNLP	NLP/CompLing	~25%	High	Best fit if NLP contribution framing dominates; different reviewer pool
Journal of Econometrics	Econometrics	~20%	High	If theoretical identification contribution is central

SPECIFIC TASKS	DEFINITION OF DONE	JOURNAL-GRADE PITFALLS
• Select target journal based on the paper's primary contribution framing: finance-first (JFE/RFS) vs. NLP-first (ACL/EMNLP). The same paper requires meaningfully different framing for each audience. • Format manuscript to journal specifications: anonymize for double-blind review (remove author names, institutional affiliations, and acknowledgments from the main file). • Write a cover letter: (1) title and one-sentence summary, (2) why this paper belongs in this journal specifically, (3) suggested associate editors by research area, (4) list of authors with conflict-of-interest disclosures. • Submit via journal's online system; retain submission confirmation and manuscript ID. • Upon receiving referee reports: read all reports fully before responding; categorize each comment as (a) easy fix, (b) requires new analysis, (c) conceptual disagreement. Respond to every single comment in a numbered response letter. • Resubmission response letter format: for each referee comment, quote the comment verbatim, provide a point-by-point response, and cite the specific page/line of the revised manuscript where the change appears. • If desk rejected: revise based on any available editorial feedback and resubmit to the next journal on the target list within 4 weeks.	• Manuscript submitted to target journal with confirmation receipt. • If revision requested: full response letter is complete, every referee comment is addressed, and revised manuscript is within journal page limits. • Replication package is publicly archived (Zenodo, GitHub, or journal dataverse) upon conditional acceptance.	• Submitting simultaneously to multiple journals: simultaneous submission is an academic ethics violation and grounds for permanent blacklisting at any journal that discovers it. • Antagonistic response letters: referees volunteer their time. Dismissive or combative responses to comments—even incorrect ones—reliably produce rejection on second review. Disagree respectfully and with evidence. • Failing to respond to every comment: referees notice when a point-by-point response skips items. This signals either carelessness or an inability to address the comment, both of which produce rejections. • Abandoning after one rejection: the first submission is a calibration exercise. Near-threshold rejections at JFE frequently become acceptances at RFS or JFQA with targeted revisions. Persistence, not perfection, characterizes published research.

Critical Risk Register: What Kills Papers at Top Journals

The following table summarizes the highest-impact risks across all phases. These are the issues that cause outright rejection—not requests for revision—at JFE, RFS, and JF. Address each proactively before submission.

Risk	Severity	Description	Mitigation
Look-Ahead Bias	FATAL	Using any post-release price, label, or model information in pre-release feature construction. Includes timezone errors in label matching.	Strict temporal splits; null-model sanity check; independent audit of label construction.
Train/Test Contamination	FATAL	Any normalization, PCA, or feature scaling fitted on the full dataset rather than training data only.	All transforms fitted on train split exclusively; documented in code with assertions.
P-Hacking / Multiple Testing	FATAL	Reporting only the best-performing model variant, subset, or window without correction for the number of tests run.	Pre-register all hypothesis tests; apply BH correction; report all results including failures.
Macro Surprise Confound	MAJOR	Model predicts yields because FOMC releases co-occur with macro data surprises, not because of text content.	Include Bloomberg consensus surprise controls in multivariate regression; run placebo on non-data-release FOMC events.
Overfitting to Policy Regime	MAJOR	Model trained on 2010–2018 low-volatility communication style fails on 2022–2024 tightening language.	Walk-forward CV with regime-spanning folds; explicitly test across tightening vs. easing sub-periods.
Missing Replication Package	MODERATE	No code or data package; reviewer cannot reproduce tables.	GitHub repository + Zenodo DOI with complete pipeline before submission.
Contribution Overclaim	MODERATE	Claiming the model 'solves' central bank communication interpretation or is 'universally superior'.	Precisely scope the contribution; acknowledge limitations; compare only to published baselines.

End of Roadmap · 5 Phases · 12 Milestones · Full Lifecycle